

EGR-338

Microcontrollers in Smart Systems

LABORATORY 1.19

Week 1

Thursday January 10, 2024

Due Thursday January 17, 2024

Names:

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FLASHING LED's BOOLEAN LOGIC OPERATIONS LOGIC

Resources:

- **Functioning C-Code Program, Bb**
“Flashing LED’s”
- **Functioning C-Code Program Logic AND Boolean Operation, Bb**
“Laboratory 1.9 Week 1 Combinational Logic”
- ***Essential-C*, Bb**

Procedure:

Part 1) Demonstrate Bread Board wiring, PICKit 3, MPLAB X IDE and provided C-Code *“Flashing LED’s”* for Sign-Off by Instructor.

Part 2) A Functioning C-Code Program is provided for you on the Desktop. The provide Program is commented and performs a Logic AND Boolean Operation. You are required to make the required jumper connections, compile and download C-Code to PIC Microcontroller, and complete the Truth Table by demonstrating to Instructor or Laboratory Assistant using LED’s on your Bread Board for Sign-Off.

Important Note: You will need to wire your Switches as Active High configuration.

Important Note: You will need to select and assign your own PIN for use as inputs and outputs on the PIC16F887 in C-Code. That means modifying the provided Functioning C-Code *“Laboratory 1.10 Week 2 Combinational Logic”*.

Download Microchip Software:

~~MPLAB X IDE v5.30~~

<https://www.microchip.com/mplab/mplab-x-ide>

~~MPLAB XC8 Compiler v2.10~~

<https://www.microchip.com/mplab/compilers>

Deploy MPLAB X IDE

Create New Project

Microchip Embedded

Standalone Project

Select device

16F887

Select Tool

PICKit 3

SN:BURxxxxxxxxx

Select Compiler

XC8

XC8v2.10)[c:\ProgramFiles(x86)\microchip\xc8\v1.35\bin]

Project Name

LED

New File *“You will find this at the Header”*

Choose File Type

C

C Source File

LED

“At this time you should see LED.c under Project at the Header”

“Type or copy template counter program provided”

Double click on LED in the upper left hand corner

Right click on Source Files

Add Existing Item

Select LED

Debug Main Project *“You will find this at the Header”*

“Look for Build Successful, counter program should now be running on Mechatronics Board”

“Look for green light Status on PICKit 3 In-Circuit Debugger”

0) Download Flashing LED's Functioning C-Code and Demonstrate on Bread Board for Signoff by Instructor.

PIN Assignments:

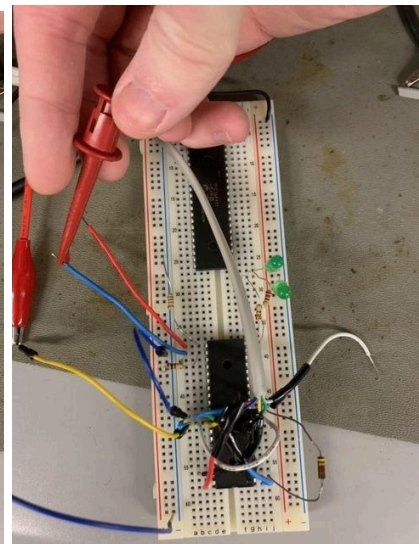
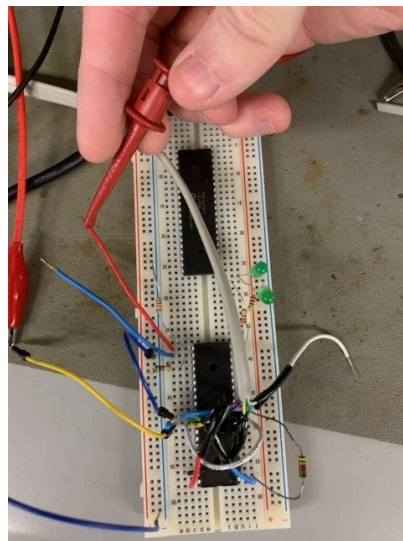
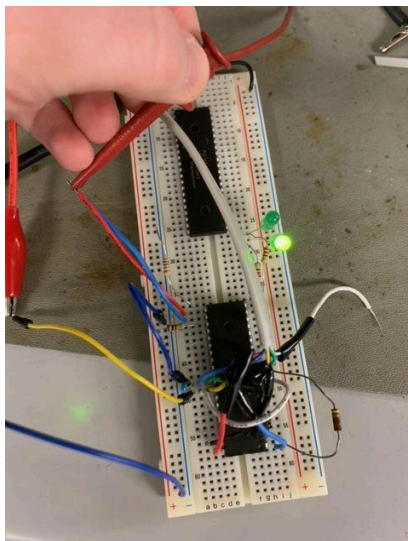
PIN RD0 Output to LED

PIN RD1 Output to LED

Instructor Signoff _____

AND Boolean Operation AND Truth Table:

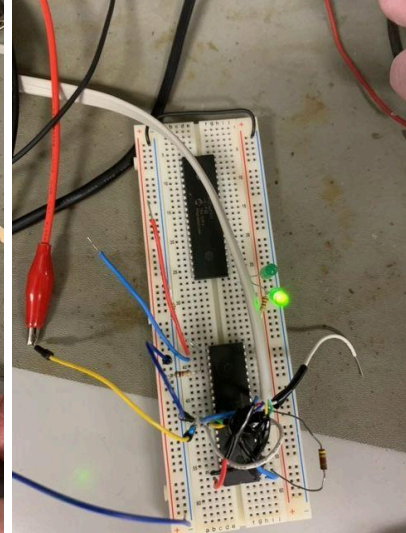
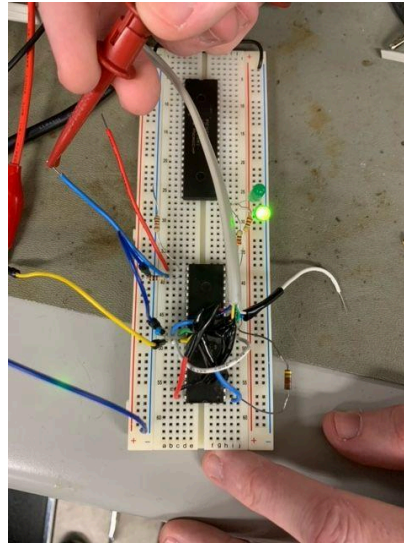
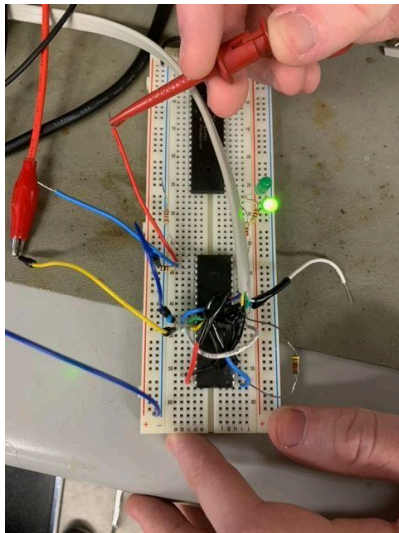
LED D0	LED D1	LED D7
0	0	0
0	1	0
1	0	0
1	1	1



1) Make required modifications to C-Code and demonstrate Logic OR Boolean Operation and complete the Truth Table by demonstrating to Instructor or Laboratory Assistant using LED's on the Mechatronics Development Board for Sign-Off

OR Boolean Operation OR Truth Table:

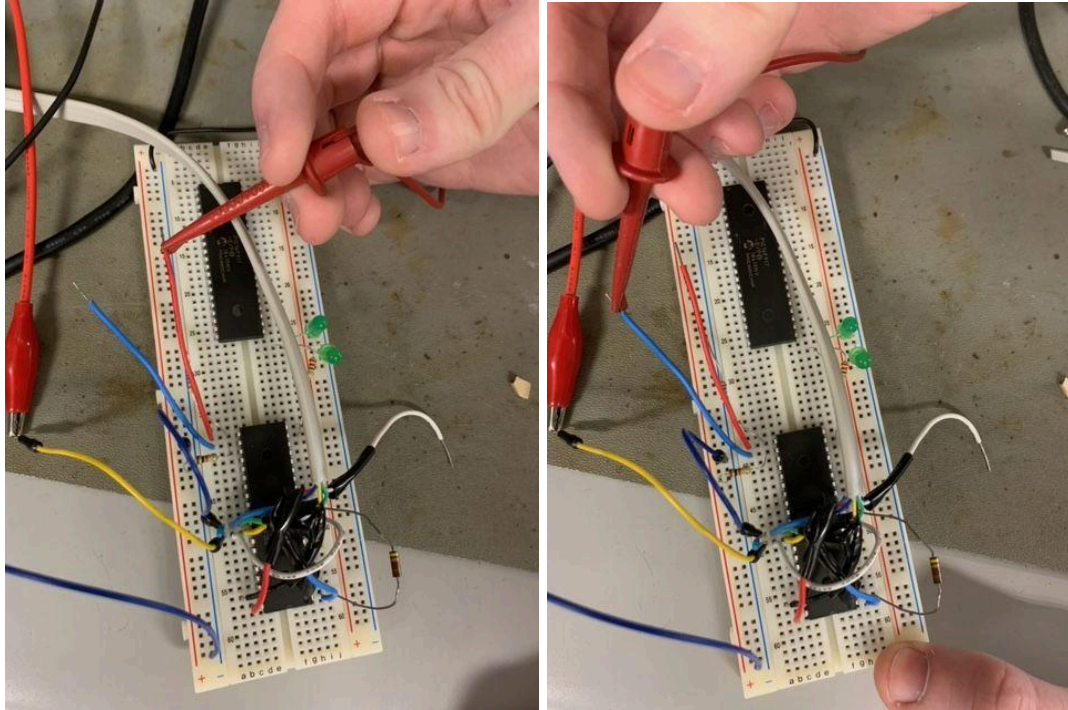
LED D0	LED D1	LED D7
0	0	0
0	1	1
1	0	1
1	1	1



2) Make required modifications to C-Code and demonstrate Logic XOR Boolean Operation and complete the Truth Table by demonstrating to Instructor or Laboratory Assistant using LED's on the Mechatronics Development Board for Sign-Off

XOR Boolean Operation XOR Truth Table:

LED D0	LED D1	LED D7
0	0	0
0	1	1
1	0	1
1	1	0



3) Make required modifications to C-Code and demonstrate Logic AND Boolean Operation for Three Inputs and complete the Truth Table by demonstrating to Instructor or Laboratory Assistant using LED's on the Mechatronics Development Board for Sign-Off.

3 Input AND Truth Table:

LED D0	LED D1	LED D2		LED D7
0	0	0		0
0	0	1		0
0	1	0		0
0	1	1		0
1	0	0		0
1	0	1		0
1	1	0		0
1	1	1		1

3-Input AND Boolean Operation

```

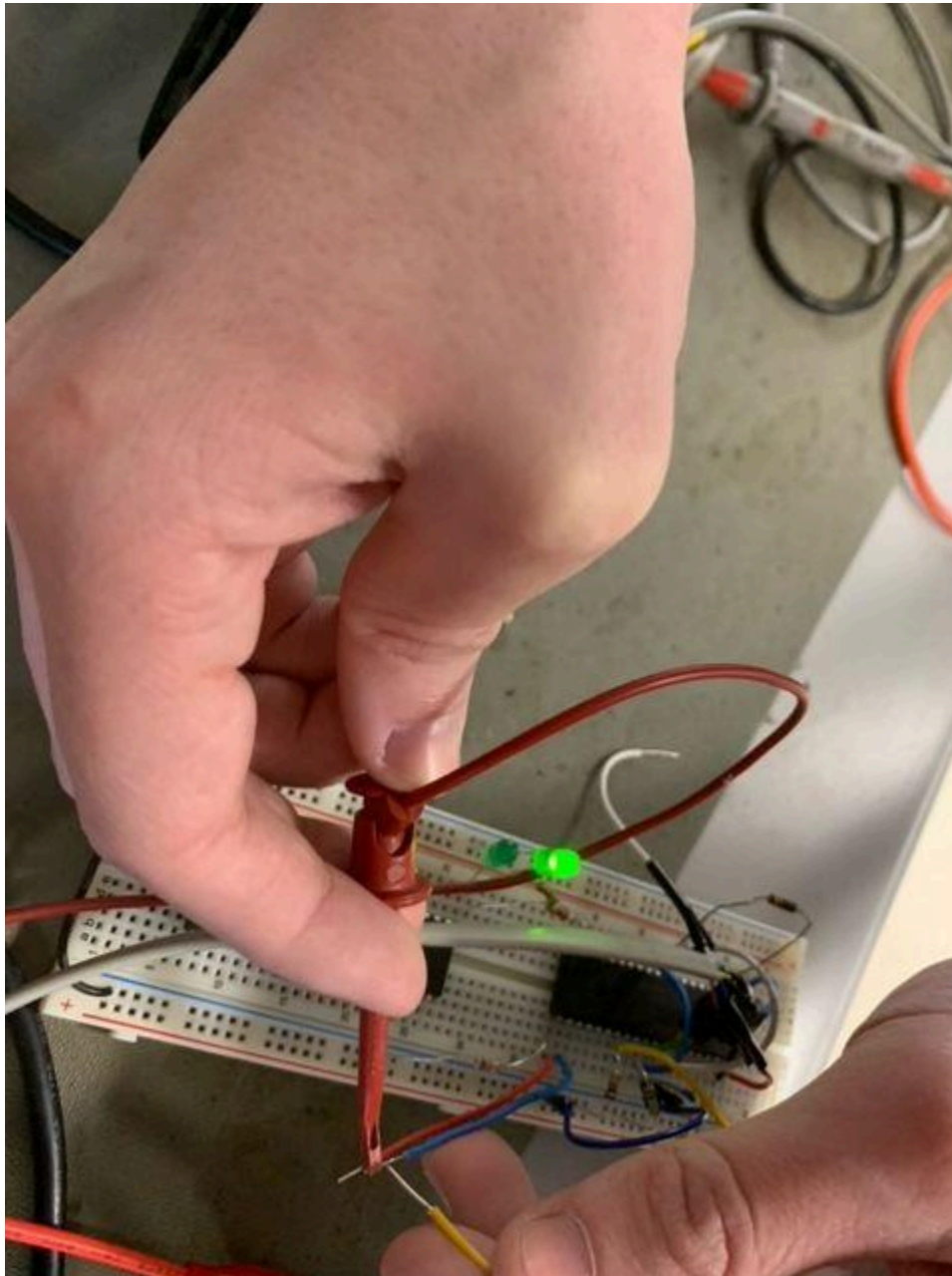
if (((RD2)&&(RD3))&&(RD4)){ //Conditional Boolean AND Function
goto Truth;} //Jump to Truth Subroutine
}

```



```
{  
Truth:  
PORTD=0b000000001; //RD0 = LED ON  
//0000_0001 Binary, 0x01 HEX
```

4) Write the SOP Equation for the given Truth Table below, Reduce using Boolean Algebra or K-Mapping, Draw Equivalent Reduced Combinational Logic Circuit, make required modifications to C-Code, and demonstrate to Instructor or Laboratory Assistant using LED's on the Mechatronics Development Board for Sign-Off. (Attach source code to submission).



SOP Truth Table:

A	B	C	D	Y
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	0
0	1	1	1	1
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	0
1	1	0	1	1
1	1	1	0	0
1	1	1	1	1

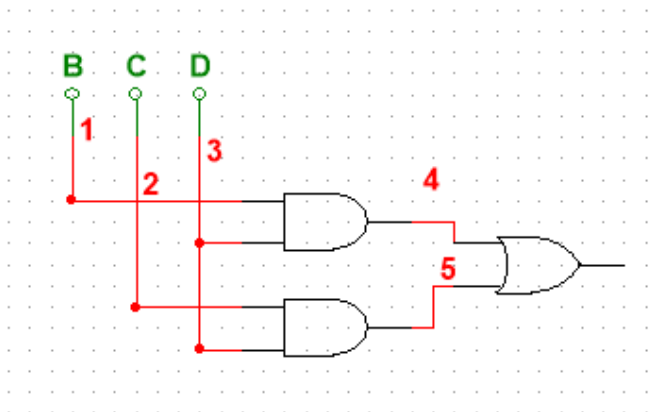
SOP Equation:

$$B'CD + BC'D + BCD$$

SOP Reduced Equation:

$$BD + CD$$

Equivalent Combinational Circuit:



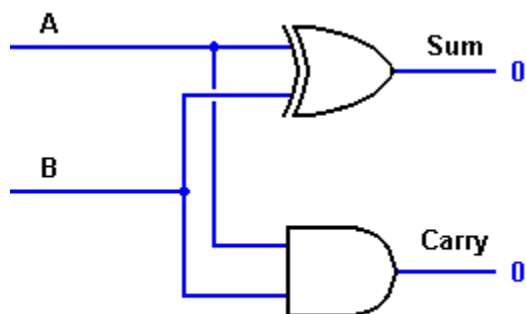
Instructor Sign Off:

Truth Table Validated in C-Code _____

5) Please make required modifications to C-Code to create a Half-Adder as seen below in the circuit, and complete truth table and demonstrate to Instructor using LED's on the Mechatronics Development Board for Sign-Off. (Attach source code to submission).

Half-Adder Truth Table:

Input A LED D0	Input B LED D1	Sum LED D6	Carry LED D7
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1



Write Boolean expressions for:

$$\text{Sum} = (A+B)'$$

$$\text{Carry} = A*B$$

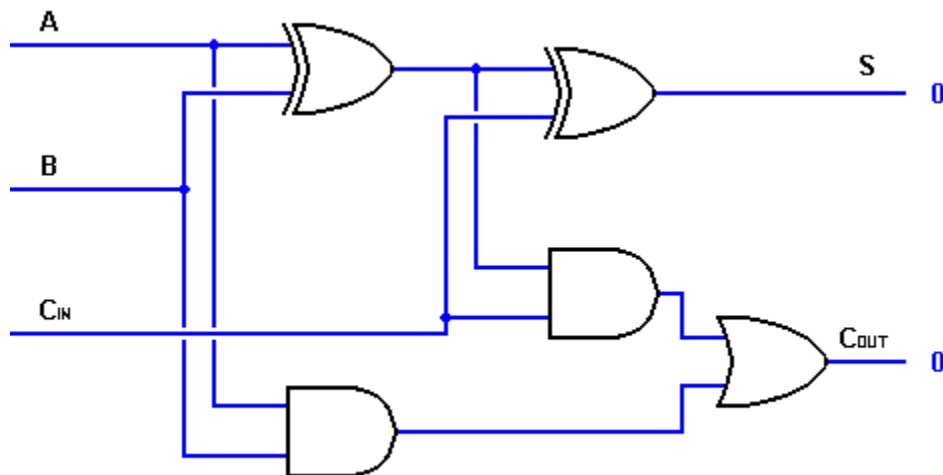
Instructor Sign Off:

Half-Adder Truth Table Validated in C-Code _____

6) Please make required modifications to C-Code to create a Full-Adder as seen below in the circuit and complete truth table and demonstrate to Instructor using LED's on the Mechatronics Development Board for Sign-Off. (Attach source code to submission).

Half-Adder Truth Table:

Input A LED D0	Input B LED D1	CarryIn LED D2	Sum LED D6	CarryOut LED D7
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1



Write Boolean expressions for:

$$\text{Sum} = ((A+B)' + \text{Cin})'$$

$$\text{Carry Out} = (A*B) + (\text{Cin}*(A+B)')$$

Instructor Sign Off:

Half-Adder Truth Table Validated in C-Code _____

Breadboard Operation Pics:

